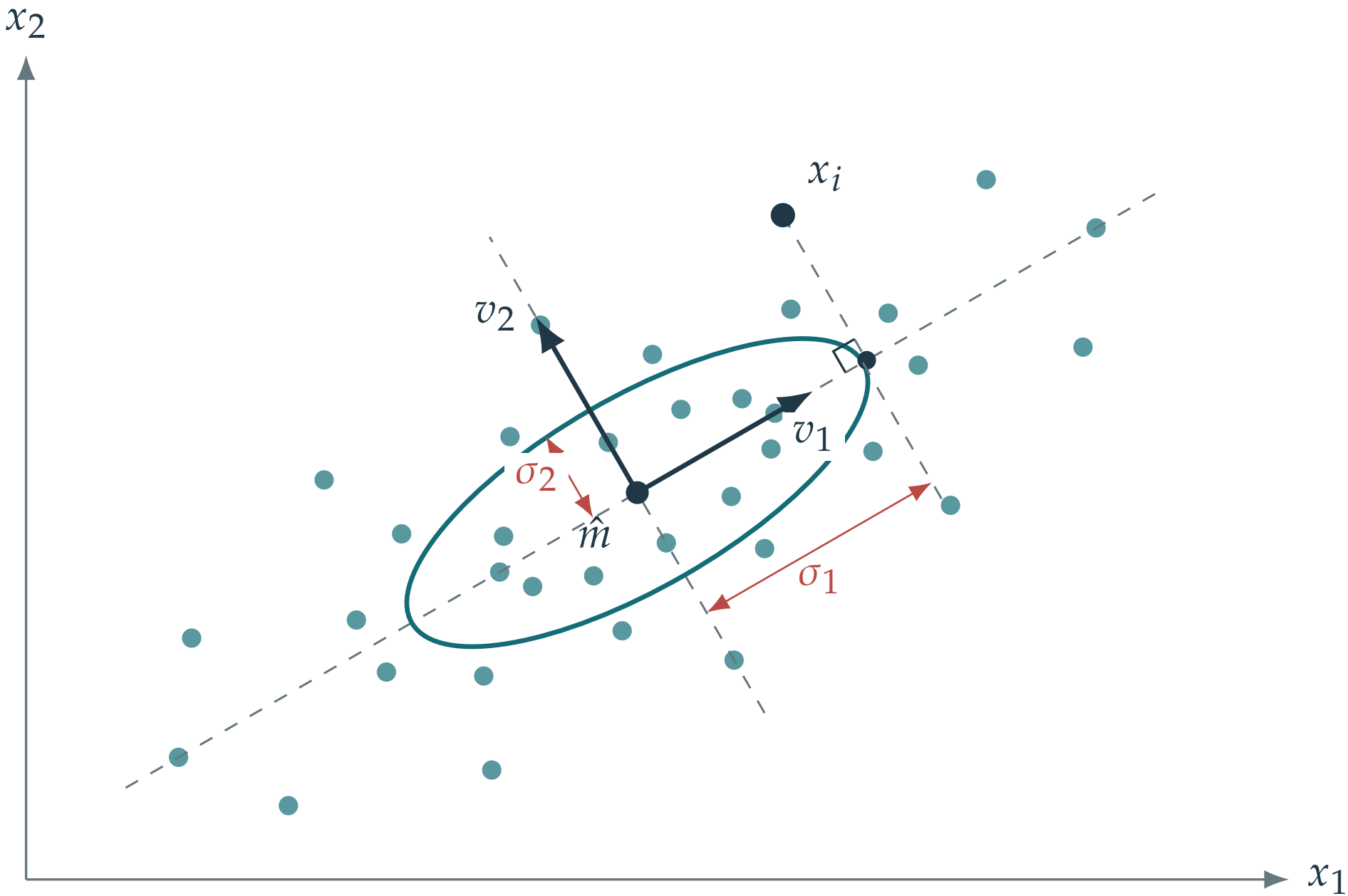


$$\widehat{C} v_j = \sigma_j^2 v_j, \qquad \|v_j\| = 1$$

$$\frac{1}{n} \sum_i \langle x_i - \hat{m}, v_j \rangle^2 = \sigma_j^2$$

Project first; the variance is the mean squared coordinate along that direction.